

The angle: comparing and measuring

Two rays from one point — and the protractor that turns them into a number.

LESSON 5–6

2 × 40 MIN

MATEMATIKA 7, §3

S8 5

LESSON CLOCK

15'

25'

25'

20'

5'

What an angle is	15 min
Measuring	25 min
The five kinds	25 min
Adding angles	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define an angle, its vertex and its arms, and use the three-letter notation.
- Measure and draw angles with a protractor.
- Classify angles as acute, right, obtuse, straight or reflex.
- Use the additive property and the bisector of an angle.

TEXTBOOK

National	pp. 15–21
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01

Explanation

What an angle is

An **angle** is the figure formed by two rays with a common origin. The origin is the **vertex** and the rays are the **arms**.

NOTATION	MEANS
$\angle ABC$	vertex B , arms through A and C
$\angle B$	the same, when no confusion is possible
$\angle 1, \angle \alpha$	a label written inside the angle

THE VERTEX LETTER GOES IN THE MIDDLE

$\angle ABC$ and $\angle CBA$ are the same angle; $\angle BAC$ is a different one. Reading the middle letter first is the habit to build.

Measuring

The unit is the **degree**: a full turn is divided into 360 equal parts. A protractor is a half-turn of 180 of them.

STEP	WHAT TO DO
1	put the centre of the protractor on the vertex
2	lay the zero line along one arm
3	read where the other arm crosses the scale
4	choose the scale that starts at 0 on your arm

Smaller units exist: $1^\circ = 60'$ (minutes) and $1' = 60''$ (seconds), used in astronomy and surveying.

A PROTRACTOR HAS TWO SCALES

One runs 0 to 180 left to right, the other right to left. Reading 130° when the angle is plainly acute means the wrong scale was used — and a glance at the figure catches it.

The five kinds

NAME	SIZE	EXAMPLE
acute	$0^\circ < \alpha < 90^\circ$	45°
right	$\alpha = 90^\circ$	a corner of a page
obtuse	$90^\circ < \alpha < 180^\circ$	130°
straight	$\alpha = 180^\circ$	a line through the vertex
reflex	$180^\circ < \alpha < 360^\circ$	250°

Two angles are **complementary** if they add to 90° and **supplementary** if they add to 180° .

COMPLEMENTARY IS THE SMALLER PAIR

C comes before S in the alphabet, and 90 before 180. That is the whole mnemonic, and it prevents the one confusion this vocabulary causes.

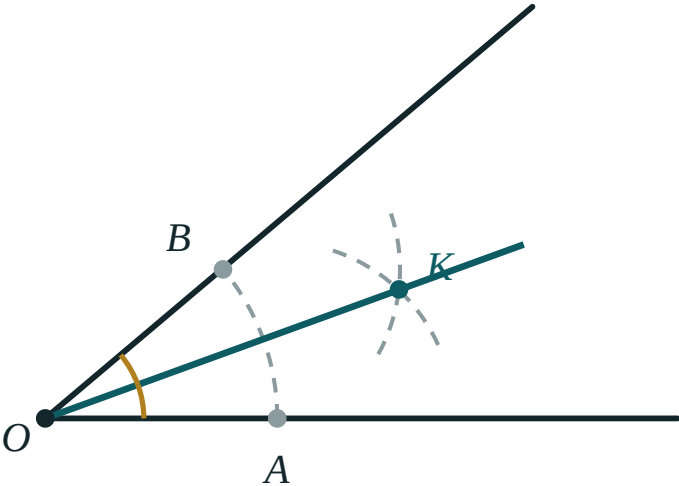
Adding angles

If the ray $[BD)$ lies inside $\angle ABC$, then

$$\angle ABC = \angle ABD + \angle DBC$$

The **bisector** of an angle is the ray from the vertex that divides it into two equal angles:

$$\angle ABD = \angle DBC = \frac{\angle ABC}{2}$$



The bisector divides the angle into two equal parts.

GIVEN	FIND	WORKING
$\angle ABC = 80^\circ$	half	40°
$\angle ABD = 35^\circ, \angle DBC = 48^\circ$	$\angle ABC$	83°
$\angle ABC = 140^\circ, \angle ABD = 55^\circ$	$\angle DBC$	85°

THE SAME ADDITIVE LAW AS FOR SEGMENTS

A point between the ends splits a segment; a ray inside the angle splits the angle. One idea, two settings — which is why the two lessons sit side by side.

02 Worked examples

EXAMPLE 1 $\angle ABD = 35^\circ$ and $\angle DBC = 48^\circ$, with $[BD]$ inside $\angle ABC$. Find $\angle ABC$.

$$1 \quad \angle ABC = \angle ABD + \angle DBC$$

$$2 \quad = 35^\circ + 48^\circ$$

$$3 \quad = 83^\circ$$

$$4 \quad \text{Acute, since } 83^\circ < 90^\circ.$$

ANSWER

83°

EXAMPLE 2 The bisector of $\angle ABC = 140^\circ$ is $[BD]$. Find $\angle ABD$, and classify both angles.

$$1 \quad \angle ABD = \frac{140^\circ}{2}$$

$$2 \quad = 70^\circ$$

$$3 \quad \angle ABC = 140^\circ \text{ is obtuse.}$$

$$4 \quad \angle ABD = 70^\circ \text{ is acute.}$$

ANSWER

70° ; obtuse and acute

EXAMPLE 3 Find the complement and the supplement of 37° .

① Complement: $90^\circ - 37^\circ$.

② $= 53^\circ$

③ Supplement: $180^\circ - 37^\circ$.

④ $= 143^\circ$

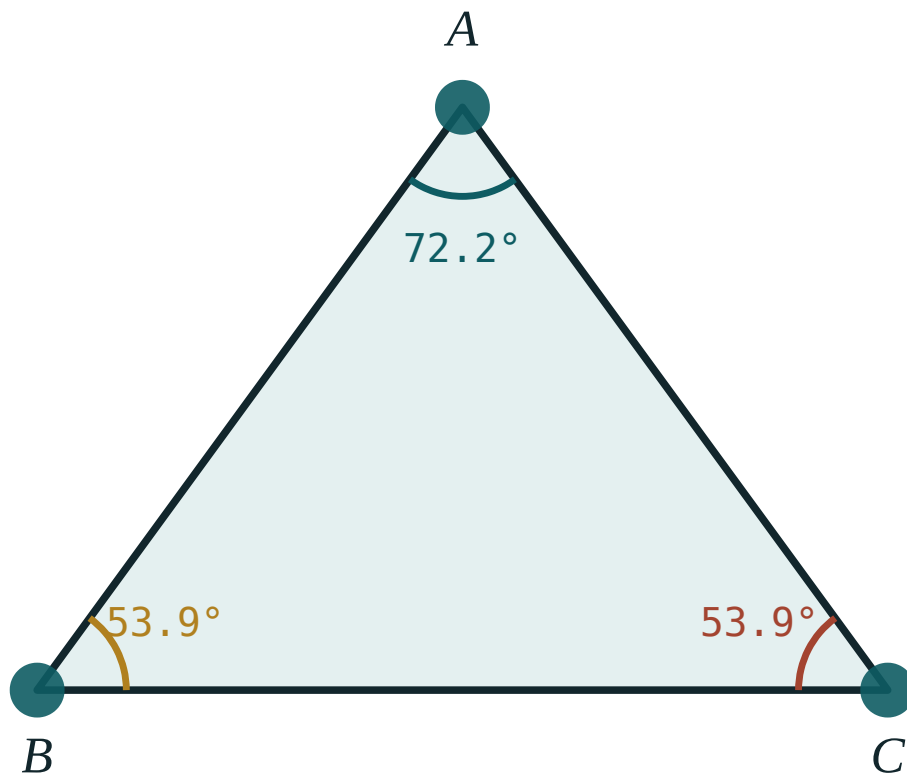
ANSWER

53° and 143°

03 Interactive model

Open a pair of scissors slowly and ask the class to call out “acute”, “right”, “obtuse”, “straight”; the classification becomes a reflex before it is a table.

Measuring and classifying

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Angle	Burchak	Угол
Vertex	Uchi	Вершина
Arm of an angle	Burchak tomoni	Сторона угла
Protractor	Transportir	Транспортир
Acute angle	O'tkir burchak	Острый угол
Obtuse angle	O'tmas burchak	Тупой угол
Straight angle	Yoyiq burchak	Развёрнутый угол
Bisector	Bissektrisa	Биссектриса

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $\angle ABC$ the vertex is:

A right angle is:

130° is:

Complementary angles add to:

90°

180°

270°

360°

The supplement of 37° :

53°

143°

323°

37°

A bisector divides an angle into:

two equal parts

three parts

two unequal parts

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify 45°

ANSWER Acute

2. Classify 90°

ANSWER Right

3. Classify 130°

ANSWER Obtuse

4. Classify 180°

ANSWER Straight

5. Classify 250°

ANSWER Reflex

6. Complement of 37°

ANSWER 53°

7. Supplement of 37°

ANSWER 143°

Medium · the standard question

7 PROBLEMS

1. $\angle ABD = 35^\circ$, $\angle DBC = 48^\circ$: $\angle ABC$

ANSWER 83°

2. $\angle ABC = 140^\circ$, $\angle ABD = 55^\circ$: $\angle DBC$

ANSWER 85°

3. Bisector of 140°

ANSWER 70° each

4. Bisector of 96°

ANSWER 48° each

5. Complement of 64°

ANSWER 26°

6. Supplement of 118°

ANSWER 62°

7. An angle equal to its own supplement

ANSWER 90°

Hard · stretch and reason

7 PROBLEMS

1. An angle is 3 times its complement

ANSWER 67.5°

2. An angle is 4 times its supplement

ANSWER 144°

3. An angle exceeds its complement by 20°

ANSWER 55°

4. Two supplementary angles in the ratio 4 : 5

ANSWER 80° and 100°

5. The angle between the bisectors of two adjacent supplementary angles

ANSWER 90°

6. The angle between the hands of a clock at 3:30

ANSWER 75°

7. At 9:20

ANSWER 160°

07 Homework

Homework — 5 tasks

Measure every drawn angle with a protractor and write the reading beside it.

1. Classify 28° , 90° , 155° and 300° .
2. Find the complement and the supplement of 54° .
3. $[BD]$ lies inside $\angle ABC = 118^\circ$ with $\angle ABD = 47^\circ$. Find $\angle DBC$.
4. An angle is twice its complement. Find it.
5. Find the angle between the hands of a clock at 4:00 and at 7:30.